



LCDs will be thinner, lighter and consume less power than today's LCD.

But just like other new technologies that have already been developed but are waiting for mass acceptance, this technology will also need large quantities of key material and fabrication plants. The process will take time and lots of convincing—and much interaction with vendors—to adopt this technology.

There are two types of technologies in the OLED industry—small-molecule and polymer. Industry insiders say it will take another 10 years for this technology to become a serious threat to the LCD market. The number of steps involved in manufacturing the LCD is much more than it is in with polymer display. This ease of manufacturing translates directly to increased yield. This will help polymer displays compete with LCD in yield basis, and therefore it is just the matter of time before other manufacturers start adopting this technique.

But even if this technology takes off, past investments in LCDs will not go down the drain—there will be improvement in this technology as well. The current display market stands at \$50 billion and this will go up to \$85 billion over the next two years. This means there's space for new technology. In around five years, we will know if this new technology finds acceptance and whether we can buy products based on this technology, off the shelf.

There are many other technologies in the making such as Ferro Electric Display, Digital Light Processing (Micro Mirror, D-ILA-Direct/Digital Image Light Amplifier) which promise to be superior to the current technology. Some have already been implemented in a small way, whereas others are still in the development phase.

Companies such as Philips are working in the area of flexible displays, which will revolutionise the whole concept of interacting

with gadgets. Before long, mobile gadgets may allow you to carry information with you and access it wherever you want. Flexible displays can be wrapped on your sleeve and you can read the news from your favourite Web site, or spread it on the table to show friends a movie from your collection.

There are many ways to achieve the goal of making flexible displays. Silicon on plastic and polymer electronics are two ways to achieve this flexibility. To create an active matrix on plastic there are a couple of ways, the most common being transferring existing silicon process from

glass to plastic. Some prototypes using this method have already been unveiled by Philips, but a major hindrance in transferring the silicon technology to the polymer substrate needs to be addressed, so it may take a while before the technology becomes commercially viable.

On the CRT front, prices are expected to drop further, though there doesn't seem to be any point in manufacturing screens larger than 22-inch. The reason could be the huge footprint of the monitor, and the very small market for displays bigger than 19-inch. Gamers and designers are not using displays bigger than 19-inch because current graphics cards don't perform any better at resolutions higher than 1,600 x 1,200 pixels.

As far as technology improvements go, CRTs are more or less the same as they were a year back and there doesn't seem to be any significant change. Video bandwidth has improved, so even 15-inch monitors are capable of displaying 1,024 x 768 x 32 at a refresh rate of 75 Hz. Dot pitch has improved too, and now some monitors sport dot pitches as small as 0.22 mm.

To sum up, CRTs don't look as hot as LCDs and plasma displays do. While introducing the new iMac with its flat-panel monitor, Apple's Steve Jobs even spoke about "the death of CRT".



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